#Probability, #Statistics

# Notebook

Day - 3

# Conditional probability.

## 1) Binomial Coefficients.

unordered samples $r$ out of $n$ without replacement

$$\binom{n}{r} \quad ({}^{n}C_{r})$$

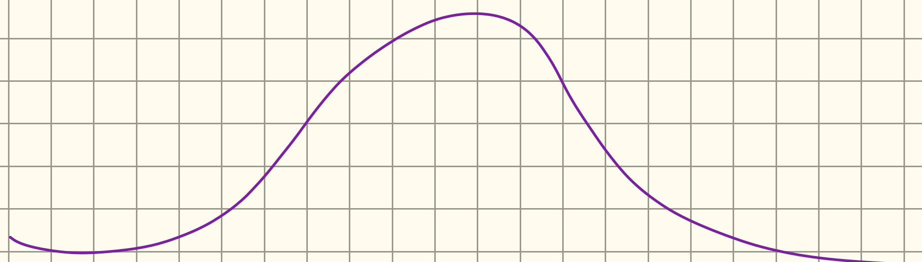
$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Pascal's triangle:

$$\begin{array}{ccccccccccc} & & & & & 1 & & & & & \\ & & & & 1 & & 1 & & & & \\ & & & 1 & & 2 & & 1 & & & \\ & & 1 & & 3 & & 3 & & 1 & & \\ & 1 & & 4 & & 6 & & 4 & & 1 & \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$



## 2) Multinomial Distribution:

$n$ Objects divided into $r$ classes of size $n_1, n_2, \ldots n_r$ and order not important.

$$\binom{n}{n_1, n_2, n_3, \dots n_r} = \frac{n!}{n_1!\, n_2! \dots n_r!}$$

$\Downarrow$

not product.

assumes that $(n_1+n_2+\dots n_r = n)$.

Proof:
$$\binom{n}{n_1\, n_2 \dots n_r} = \binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3} \dots \binom{n-n_1-n_2 \dots - n_{r-1}}{n_r}$$

$$= \frac{n!}{\cancel{(n-n_1)!}\, n_1!} \times \frac{\cancel{(n-n_1)!}}{\cancel{(n-n_1-n_2)!}\, n_2!} \times \dots \times \frac{\cancel{(n-n_1-n_2 \dots n_{r-1})!}}{n_r!\, 0!}$$

$$= \frac{n!}{n_1!\, n_2!\, n_3! \dots n_r!}$$

e) 7 people make 3, 2, 2 sub-Committees:

$$\binom{7}{3,2,2} = \frac{7!}{3!\, 2!\, 2!}$$

$$= \frac{7 \times 6 \times 5 \times \cancel{4}}{\cancel{4}}$$

$$= \boxed{210}$$

ok

$$\binom{7}{3}\binom{7-3}{2}\binom{7-3-2}{2}$$

$$\frac{7!}{\cancel{4!}\,3!}\times\frac{\cancel{4!}}{2!\,\cancel{2!}}\times\frac{\cancel{2!}}{0!\,2!}$$

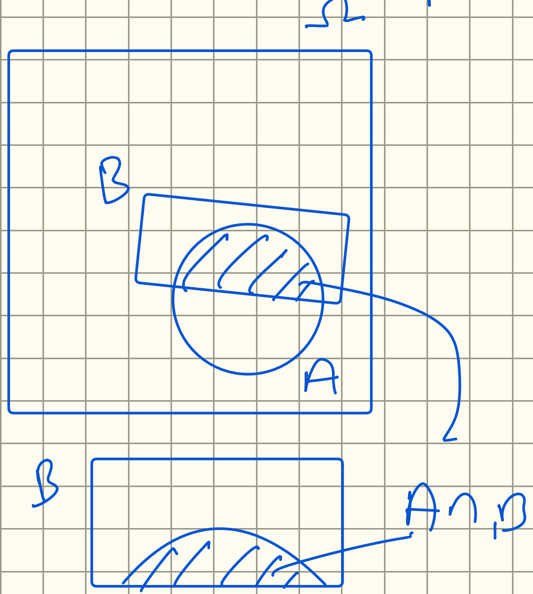
$$\frac{7!}{3!\,2!\,2!} = \boxed{210}$$

# Conditional Probability

A, B are two events (outcomes) of a random experiment.

Prob. of A given B.

$$P(A|B) = \frac{P(A\cap B)}{P(B)}$$




Ex:

A = first die = 3

B = second die = 5

C = sum dice = 6

$$P(A|B) = P(A) = \frac{1}{6}$$

$P(A \mid C) = ??$

$= \dfrac{P(A \cap C)}{P(C)} = \dfrac{\frac{1}{6} \times \frac{5}{36}}{\frac{5}{6^2}}$ $C = \{15, 24, 33, 42, 51\}$

$= \boxed{\dfrac{1}{6}}$.

Q) Test 1000 people for cancer:

500 have cancer — 450 +, 50 −

500 don't have cancer — 100 +, 400 −

A

B

$P(A \mid B) = \dfrac{\frac{450}{1000}}{\frac{500}{1000}} = \dfrac{P(A \cap B)}{P(B)}$

$\boxed{P(B \mid A) = ??}$

Bayes' Theorem

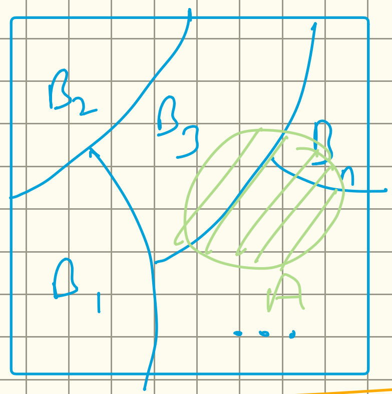
Inference.

# Law of Total probability.

$$P(A|B) = \frac{P(A\cap B)}{P(B)}$$

$$P(A\cap B) = P(A|B)\cdot P(B) \quad \text{[multiplication law].}$$

Derivation:




$$\Omega = \bigcup_{i=1}^{n} B_i$$

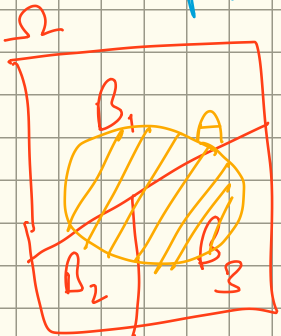
(disjoint) $B_i \cap B_j = \phi$ if $i \neq j$

$$P(A) = \sum_{i=1}^{n} P(A|B_i)\, P(B_i)$$

$$\begin{aligned} P(A) &= P(A\cap\Omega) \\ &= P\left(A\cap \bigcup_i B_i\right) \\ &= P\left(\bigcup_i (A\cap B_i)\right) \\ &= \sum P(A\cap B_i) = \sum \frac{P(A|B_i)}{P(B_i)} \end{aligned}$$

let's say for 3 B's.

$$P(A) = P(A|B_1)\cdot P(B_1) + P(A|B_2)\cdot P(B_2) + P(A|B_3)\cdot P(B_3)$$




$$= \frac{P(A\cap B_1)}{\cancel{P(B_1)}}\cdot \cancel{P(B_1)} + \frac{P(A\cap B_2)}{\cancel{P(B_2)}}\cdot \cancel{P(B_2)} + \frac{P(A\cap B_3)}{\cancel{P(B_3)}}\cdot \cancel{P(B_3)}$$

$$= P(A\cap B_1) + P(A\cap B_2) + P(A\cap B_3)$$

disjoint

$= p(A)$